

Hamed Hojatian

6G/AI Research Engineer

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Education

2019-2023	University of Montreal (Montreal Polytechnique) <i>Ph.D. in AI-aided wireless communication systems</i> Supervisors: Prof. François Leduc-Primeau, Prof. Jean-François Frigon <i>GPA: 4.0/4.0</i>
2014-2017	Isfahan University of Technology (IUT) <i>M.Sc. in communication system engineering</i> Supervisor: Prof. Mohammad Javad Omid <i>GPA: 3.7/4.0</i>
2010-2014	Babol (Noshirvani) University of Technology (NIT) <i>B.Sc. in electrical engineering</i> Supervisor: Prof. Hossein Miar-Naeimi <i>GPA: 3.1/4.0</i>

Research skills and Qualifications

- Strong proficiency in machine learning techniques such as regression, classification, clustering, and neural networks;
- Skilled in developing and implementing machine learning algorithms using a range of techniques, including supervised learning, unsupervised learning, and reinforcement learning;
- Understanding of deep reinforcement learning techniques, with experience implementing and optimizing various algorithms such as Q-learning, and deep deterministic policy gradients (DDPG);
- Experience with massive MIMO, beamforming techniques, coordinated beamforming, distributed antennas, cell-free communication and their applications in wireless communication systems, 5G beyond, and 6G;
- Experience in developing and implementing deep learning-based beamforming solutions, with a focus on optimizing beamforming performance for wireless communication systems;
- Skilled in using quantization techniques to optimize deep neural networks (DNNs) for specific hardware constraints and other use cases;
- Developed and implemented machine learning models for wireless communication systems, including channel estimation, interference cancellation, and beamforming;
- Published research papers in leading IEEE journals conferences;
- Collaborated with cross-functional teams to identify research opportunities and develop solutions that improved the reliability, efficiency, and scalability of wireless communication systems;
- Mentored graduate and undergraduate students on research projects and provided guidance on technical skills and research methodologies;

Skills

Tools and Languages | Python, Pytorch, C++ (intermediate), MATLAB, and Microsoft Office.

Languages | English, French (basic), Persian (native)

Research Experience

Ph.D.	AI-aided Beamforming in Massive MIMO Systems • Designing hybrid beamforming techniques for wireless networks, including an innovative RSSI-based hybrid beamforming design that leverages deep learning algorithms to enhance spectral and energy efficiency; • Developing unsupervised learning techniques for intelligent beamforming and coordinated hybrid beamforming in cell-free massive MIMO systems; • Exploring subarray hybrid beamforming and efficient hardware design driven by unsupervised learning techniques, to reduce the complexity of wireless communication systems; • Publishing in top-tier conferences and journals in the field of wireless communications, my research showcases my expertise in developing cutting-edge technologies to improve wireless network performance; • Filing a patent for the invention in collaboration with Ericsson as a co-op research intern, I developed a novel AI-aided method for efficient massive MIMO beamforming.
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Work Experience

Nov 2023 - present	Huawei Technologies Ltd <i>Research Engineer in Advance Wireless Research LAB</i> • At Huawei, I engaged in pioneering research on 6G wireless communication, emphasizing physical layer innovations and algorithm design. My work included developing protocols for Integrated Sensing and Communication (ISAC) and localization, enhancing connectivity. Additionally, I explored AI-driven solutions to advance wireless communication and sensing technologies, contributing to the future of telecommunications.
Nov 2022 - Nov 2023	Ericsson <i>Research intern, Co-op research</i> • Energy efficiency project in Global Artificial Intelligence Accelerator (GAIA) • Conducted research on energy models, hardware impairments, and imperfect channel models to optimize designs. • Developed energy-efficient algorithms for massive MIMO hybrid beamforming, enhancing performance and sustainability. • Leveraged AI solutions to design energy-efficient massive MIMO beamforming, contributing to innovative wireless technology.
Mar 2019 - Aug 2019	University of Montreal (Montreal Polytechnique) <i>Research Assistant</i>
Nov 2015 - Nov 2018	Mobile Telecommunication Company of Iran (MCI) <i>Radio planning and optimization engineer</i> • Live network Ericsson OSS optimization and parameter tuning for 2G, 3G, and LTE; • Network Planning and optimization for 2G, 3G, and LTE, new site baseline parameters tuning and planning;

Awards

2025	"Future Star" award @ Huawei
2024	Best thesis award nomination "Beamforming design for massive MIMO systems with deep neural networks "

Mentoring

Throughout my Ph.D. studies, I had the privilege of mentoring one Ph.D., two master students, and one internship student in our research project on wireless communication systems by providing guidance and support in their research work.

Community Service

IEEE Journals reviewer	Selected Areas in Communications (J-SAC), Transactions on Wireless Communications (TWC), Transactions on Communications (TCOMM), Wireless Communications Letters (WCL), Communications Letters (CommL), and Transactions on Vehicular Technology (TVT). Details in here .
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Certificates

2022	Introduction to Unix Shell, Calcul Quebec . Massive data analysis with Spark, Calcul Quebec . Introduction to GPU Programming with Python, Calcul Quebec .
2019	4 th IVADO/MILA SCHOOL IN DEEP LEARNING .
2018	DEEP LEARNING WITH PYTORCH, Udemy . LTE OPTIMIZATION OF RADIO NETWORK (3GPP), TechCom .
2017	LTE COUNTER AND KPI (ERICSSON), TechCom . LTE RADIO NETWORK PLANNING, TechCom .

Publications

2024	1. Hasanzadeh Karkan, A., Hojatian, H. , Frigon, J.-F. & Leduc-Primeau, F. “ SAGE-HB: Swift Adaptation and Generalization in Massive MIMO Hybrid Beamforming ”. <i>Accepted IEEE International Conference on Machine Learning and Computer Networking (ICMLCN) 2024</i> (2024). 2. Hojatian, H. , Mlika, Z., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. “ Learning Energy-Efficient Transmitter Configurations for Massive MIMO Beamforming ”. <i>IEEE Transactions on Machine Learning in Communications and Networking</i> (2024).
2022	3. Hojatian, H. , Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. “ Decentralized Beamforming for Cell-Free Massive MIMO with Unsupervised Learning ”. <i>IEEE Communications Letters</i> (2022). 4. Hojatian, H. , Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. “ Flexible Unsupervised Learning for Massive MIMO Subarray Hybrid Beamforming ”. <i>IEEE GLOBECOM</i> (2022).
2021	5. Hojatian, H. , Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. “ Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming ”. <i>IEEE Transactions on Wireless Communications</i> (2021).
2020	6. Hojatian, H. , Nguyen Ha, V., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. “ RSSI-Based Hybrid Beamforming Design with Deep Learning ”. <i>IEEE ICC</i> (2020).
2016	7. Hojatian, H. , Omid, M. J., Saeedi-Sourck, H. & Farhang, A. “ Joint CFO and Channel Estimation in OFDM-based Massive MIMO Systems ”. <i>IEEE IST</i> (2016).

Patent

2023	H. Hojatian, Z. Mlika, J. Nadal, J. Frigon and F. Leduc-Primeau , “ ENERGY-EFFICIENT MASSIVE MIMO BEAMFORMING WITH MACHINE LEARNING OPTIMIZATION ”, U.S. patent, filed by Ericsson, PCT/IB2023/053214.
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Talks and Teaching Experience

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| 2022 | | Paper Presentation, IEEE GLOBECOM 2022 , Rio-de-Janeiro, Brazil.
Hybrid-Seminars on Selected Topics in Signal Processing.
Presenting my research progress in IVADO - Digital October . |
| 2021 | | Presenting my research progress in IVADO - Digital October . |
| 2020 | | Paper Presentation, IEEE ICC 2020 , Dublin, Ireland.
Presenting my research topic in IVADO - Digital October . |
| 2013 | | Teaching assistant for Electronic I .
Teaching assistant for Electronic circuit I . |
| 2012 | | Teaching assistant for Digital Signal Processing . |

Strengths

Hard-working, Decision Making, Ability to Work Under Pressure, Motivator & Leader Conflict Resolution, Time Management, Target oriented, Flexibility, and Adaptability